

PROBLEM 4.A2

A) 1) What is the actual feed tray?

The actual feed tray is $\boxed{0.60^{th}}$

2) What is the mole fraction MVC in the feed?

The mole fraction is $\boxed{z = 0.51}$

3) What is the vapor composition of the feed Tray?

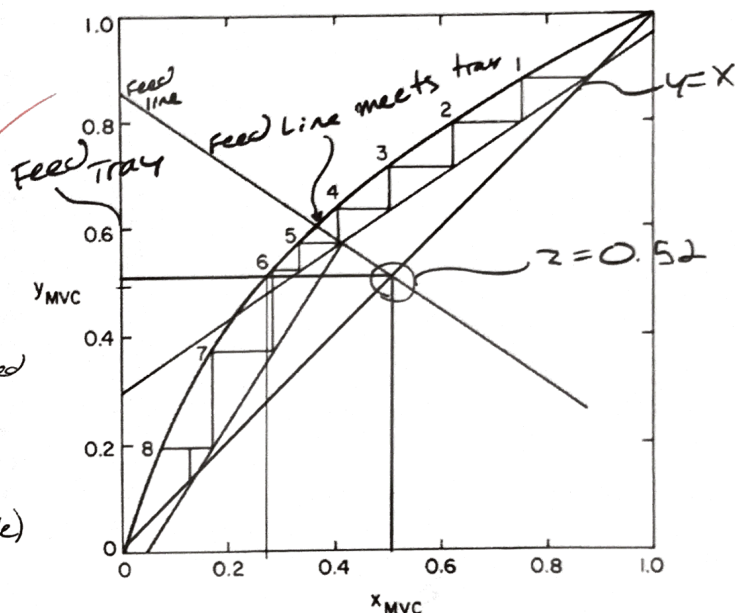
$\boxed{y = 0.51}$

4) What is the liquid composition of the feed Tray?

$\boxed{x = 0.29}$

B) Is the feed a superheated vapor feed, saturated vapor feed, two-phase feed, saturated liquid feed, or subcooled liquid feed? Looking at the slope of the feed line, the feed must be two-phase (negative slope). & we only see top & bottom O.P. Lines (NO middle)

C) Is the temperature at stage 7 higher, lower, or the same as at stage 1? Higher



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HOMEWORK #4

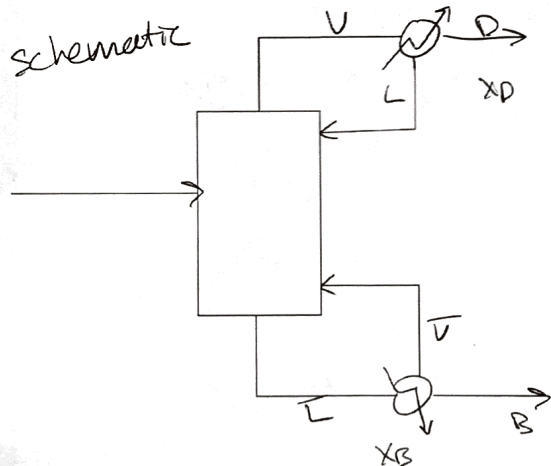
2) PROBLEM D.7

A distillation column w/ a total condenser is separating a feed that is 30.0% mol acetone from ethanol. Distillate $x_D = 0.90$ and bottoms $x_B = 0.13$ both mole fractions acetone. Assume CMO is valid. $\frac{L}{V} = 0.8$ & Boilup ratio $\frac{V}{B} = 1.0$.

A) Find the composition of the liquid leaving the fifth stage below the total condenser.

* Plot top & Bottom O.P. lines along w/ the data provided.

* Draw stages on the graph.



TOP O.P. Line: Given that $\frac{L}{V} = 0.8$ & $x = y = x_D = 0.90$

$$y = \frac{L}{V}x + (1 - \frac{L}{V})x_D$$

$$y = \frac{8}{10}x + (1 - \frac{8}{10})\frac{9}{10}$$

$$y = \frac{4}{5}x + \frac{1}{5} \cdot \frac{9}{10}$$

* TOP OPERATING LINE: $y = \frac{4}{5}x + \frac{9}{50}$

Bottom O.P. Line: $\frac{V}{B} = 1.0$, $x = y = x_B = 0.13$

$$y = \frac{V}{B}x - (\frac{V}{B} - 1)x_B$$

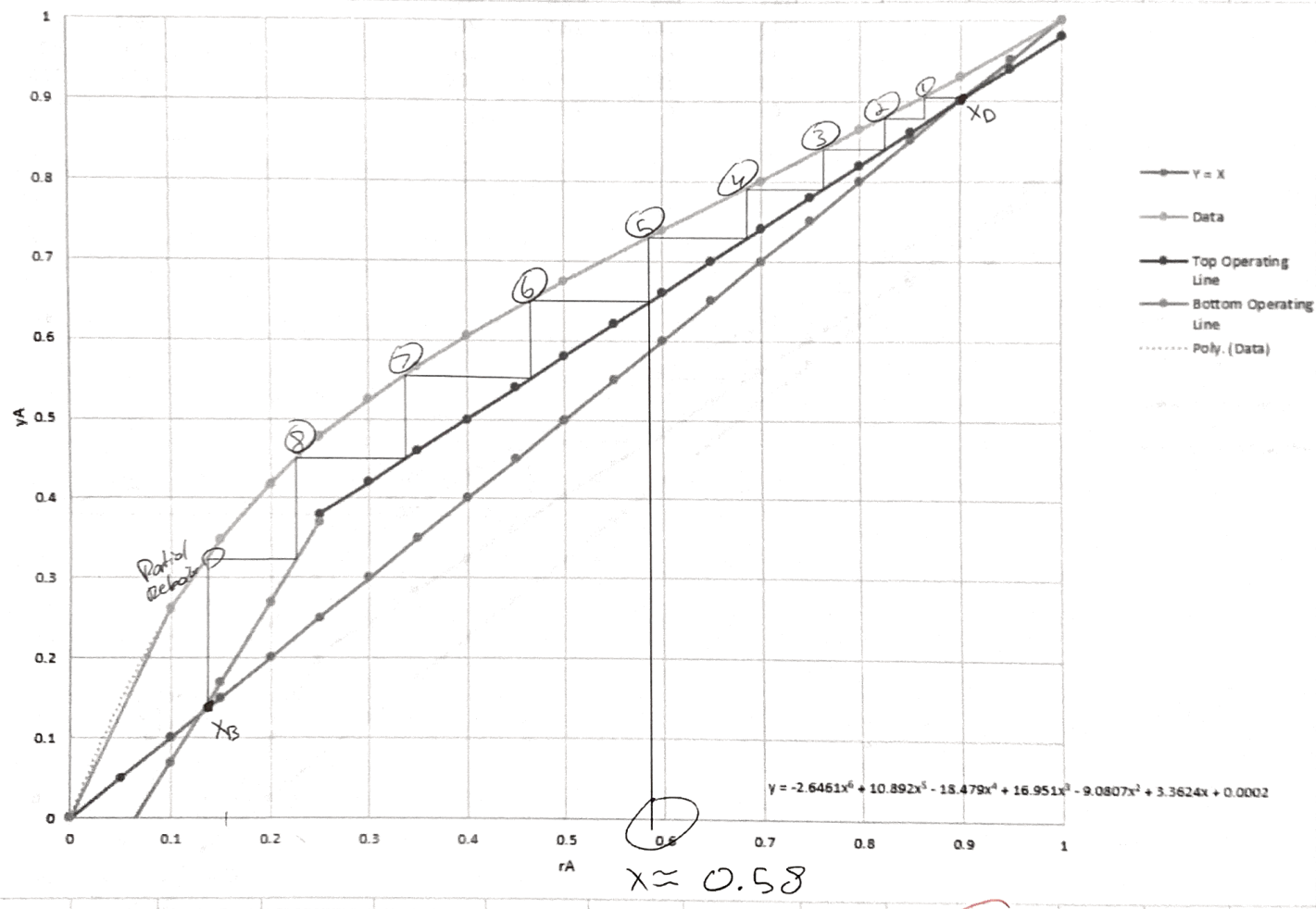
$$\text{w/ } L = V + B$$

$$y = (\frac{V}{V} + \frac{B}{V})x - (\frac{V}{V} + \frac{B}{V} - 1)x_B$$

$$y = (1 + \frac{1}{1})x - (1 + \frac{1}{1} - 1)0.13$$

* BOTTOM OPERATING LINE: $y = 2x - 0.13$

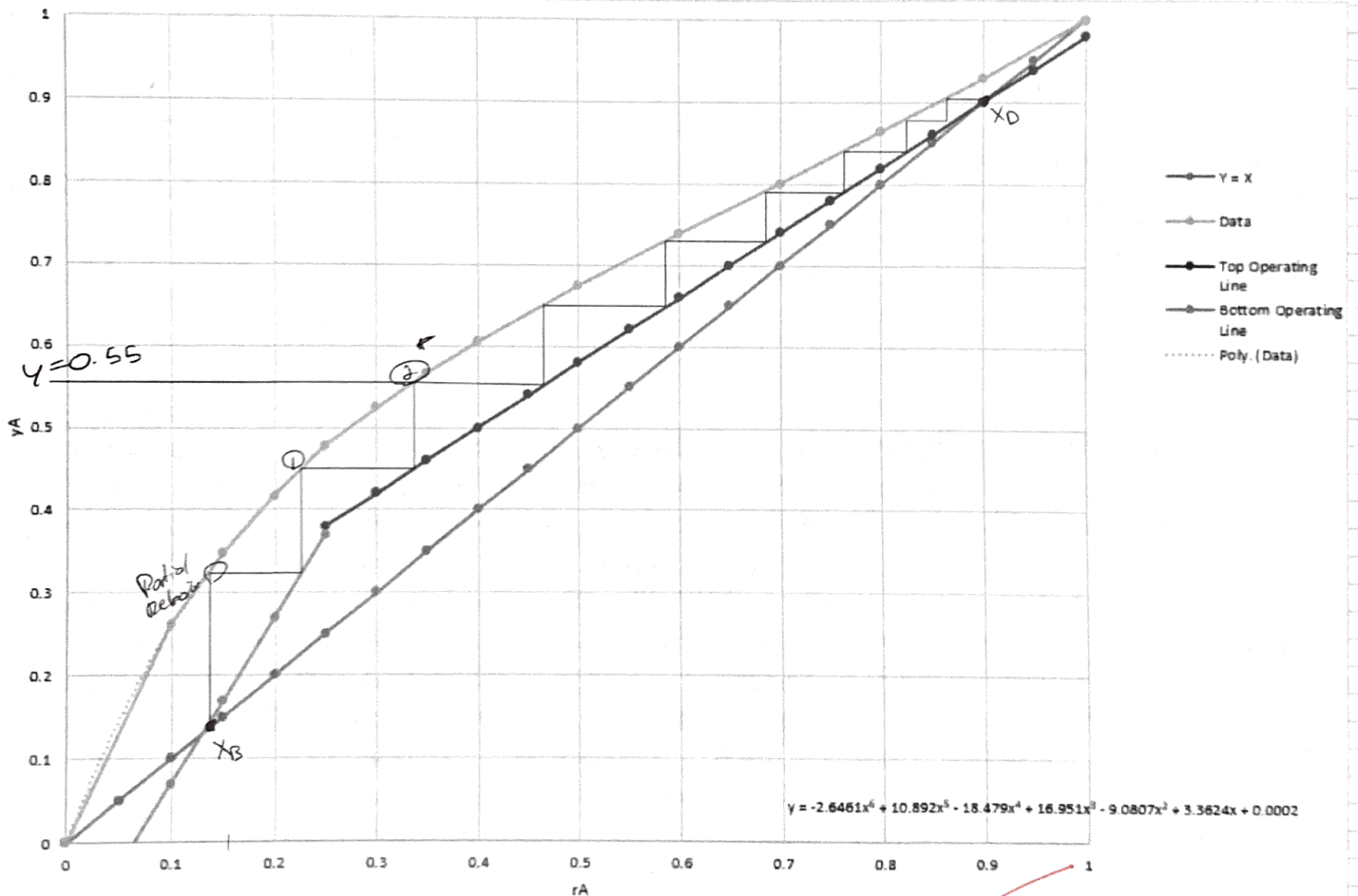
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
4 D7														
2	Y = X		Data		Top Operating Line		Bottom Operating Line							
3	X	Y	x _A	y _A	X	Y	X	Y						
4	0	0	0	0	0	0	0.18	0	-0.13					
5	0.05	0.05	0.1	0.262	0.05	0.22	0.05	0.05	-0.03					
6	0.1	0.1	0.15	0.348	0.1	0.26	0.1	0.1	0.07					
7	0.15	0.15	0.2	0.417	0.15	0.3	0.15	0.15	0.17					
8	0.2	0.2	0.25	0.478	0.2	0.34	0.2	0.2	0.27					
9	0.25	0.25	0.3	0.524	0.25	0.38	0.25	0.25	0.37					
10	0.3	0.3	0.35	0.566	0.3	0.42	0.3	0.3	0.47					
11	0.35	0.35	0.4	0.605	0.35	0.46	0.35	0.35	0.57					
12	0.4	0.4	0.5	0.674	0.4	0.5	0.4	0.4	0.67					
13	0.45	0.45	0.6	0.739	0.45	0.54	0.45	0.45	0.77					
14	0.5	0.5	0.7	0.8	0.5	0.58	0.5	0.5	0.87					
15	0.55	0.55	0.8	0.865	0.55	0.62	0.55	0.55	0.97					
16	0.6	0.6	0.9	0.929	0.6	0.66	0.6	0.6	1.07					
17	0.65	0.65	1	1	0.65	0.7	0.65	0.65	1.17					
18	0.7	0.7			0.7	0.74	0.7	0.7	1.27					
19	0.75	0.75			0.75	0.78	0.75	0.75	1.37					
20	0.8	0.8			0.8	0.82	0.8	0.8	1.47					
21	0.85	0.85			0.85	0.86	0.85	0.85	1.57					
22	0.9	0.9			0.9	0.9	0.9	0.9	1.67					
23	0.95	0.95			0.95	0.94	0.95	0.95	1.77					
24	1	1			1	0.98	1	1	1.87					



@ Stage 5 X = 0.58

13) The column has a partial re-boiler that acts as an equil. contact. Find the vapor composition leaving the second stage above the partial reboiler.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	4 D7													
2	Y = X		Data		Top Operating Line		Bottom Operating Line							
3	X	Y	x _A	y _A	X	Y	X	Y						
4	0	0	0	0	0	0.18	0	-0.13						
5	0.05	0.05	0.1	0.262	0.05	0.22	0.05	-0.03						
6	0.1	0.1	0.15	0.348	0.1	0.26	0.1	0.07						
7	0.15	0.15	0.2	0.417	0.15	0.3	0.15	0.17						
8	0.2	0.2	0.25	0.478	0.2	0.34	0.2	0.27						
9	0.25	0.25	0.3	0.524	0.25	0.38	0.25	0.37						
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11	0.35	0.35	0.4	0.605	0.35	0.46	0.35	0.57						
12	0.4	0.4	0.5	0.674	0.4	0.5	0.4	0.67						
13	0.45	0.45	0.6	0.739	0.45	0.54	0.45	0.77						
14	0.5	0.5	0.7	0.8	0.5	0.58	0.5	0.87						
15	0.55	0.55	0.8	0.865	0.55	0.62	0.55	0.97						
16	0.6	0.6	0.9	0.929	0.6	0.66	0.6	1.07						
17	0.65	0.65	1	1	0.65	0.7	0.65	1.17						
18	0.7	0.7			0.7	0.74	0.7	1.27						
19	0.75	0.75			0.75	0.78	0.75	1.37						
20	0.8	0.8			0.8	0.82	0.8	1.47						
21	0.85	0.85			0.85	0.86	0.85	1.57						
22	0.9	0.9			0.9	0.9	0.9	1.67						
23	0.95	0.95			0.95	0.94	0.95	1.77						
24	1	1			1	0.98	1	1.87						



$$y_2 = 0.55$$

c) Find the optimum feed plate location, total number of stages, and required q value of feed. $q = \frac{L}{F}$

* Feed plate line must intersect $0.3 = z = x = y$

* Feed Line Should intersect w/ the two operating line intersections :-

$$\left. \begin{aligned} y &= \frac{4}{5}x + \frac{9}{50} \text{ TOP} \\ y &= 2x - 0.13 \text{ BOTTOM} \end{aligned} \right\} \text{compute intersection}$$

$$\frac{4}{5}x + \frac{9}{50} = 2x - \frac{13}{100}$$

$$\frac{9}{50} + \frac{13}{100} = \left(2 - \frac{4}{5}\right)x \text{ plug in for } y$$

$$\frac{31}{100} \left(\frac{5}{6}\right) = x$$

$$\text{intersects @ } x = 0.258$$

$$\rightarrow y = 0.3867$$

$$\left. \begin{aligned} \text{using this} \\ x = y = z = 0.3 \end{aligned} \right\} \text{Find the slope}$$

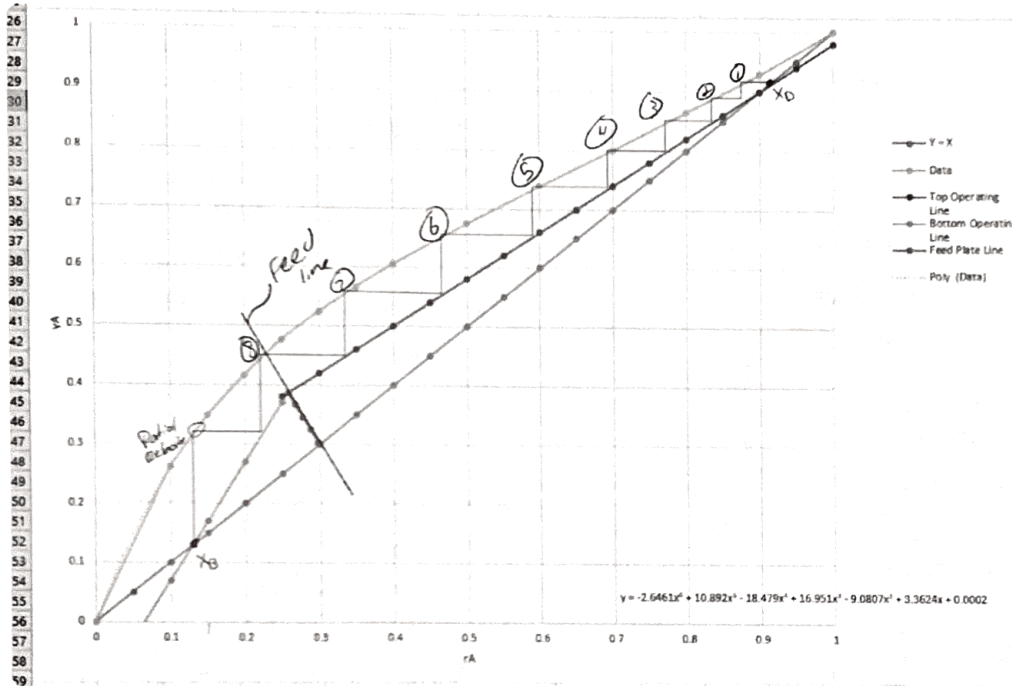
$$\text{slope} = \frac{0.3 - 0.3867}{0.3 - 0.258} = -2.08$$

y intercept

$$y = -2.08x + b$$

$$b = 0.924$$

plotting $y = -2.08x + 0.924$



Optimum location for the feed plate ↓

Between stage 7 & 8.

TOTAL: 8 STAGES & PARTIAL REBOILER

$$q = 0.675$$

$$\text{slope} = -2.08 = \frac{q}{q-1}$$

$$-2.08(q-1) = q$$

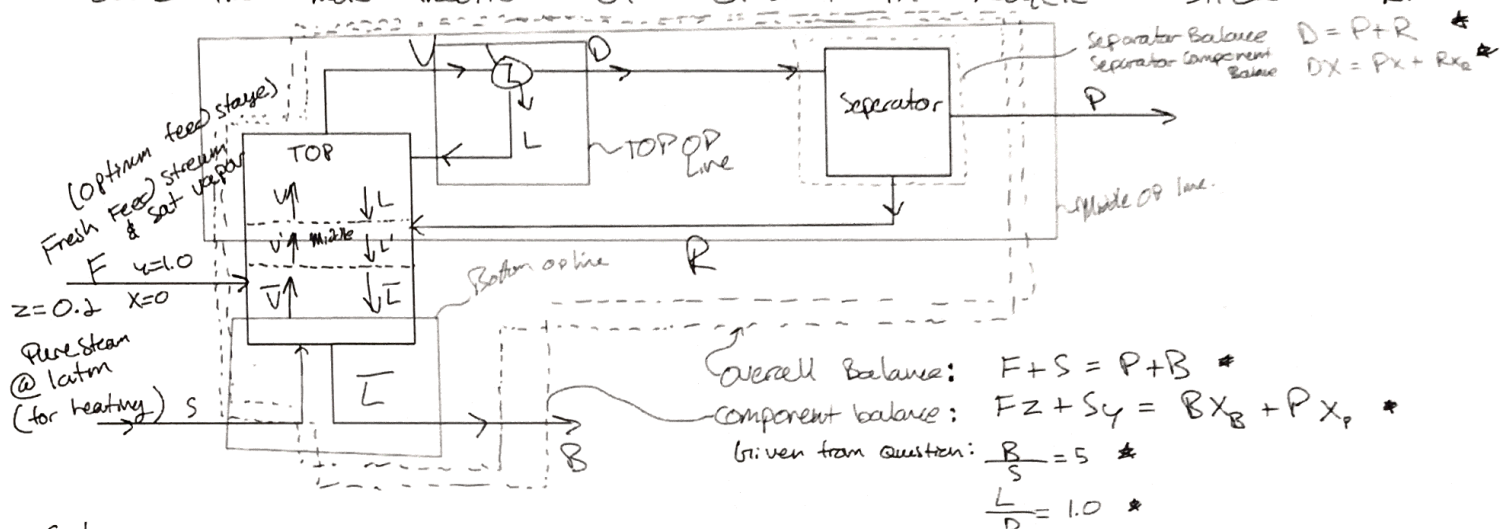
$$-3.08q = -2.08$$

$$q = 0.675$$

3) PROBLEM 4.12

A Distillation Column is connected to another separation device. The fresh feed to the column is $F = 1000.0$ kmol/h of a saturated vapor that is 20.0 mol% ethanol and 80.0 mol% water. The feed is input at its optimum feed stage. The distillation column has a partial condenser and no reboiler. The column is heated with direct injection of pure steam that is a saturated vapor at 1.0 atm. The ratio bottoms flow rate/steam flow rate $= B/S = 5.0$. The reflux ratio $L/D = 1.0$. The distillate from the column is 70.0 mol% ethanol and the bottoms is 1.0 mol% ethanol. The column operates at a pressure of 1.0 atm, the reflux is returned as a sat. liquid, and CMO is assumed to be valid. The distillate is sent to a separation device that, because it operates on a different principle than distillation, is not limited by the presence of an azeotrope. The device produces product P and and recycle stream R that is a saturated liquid returned to the distillation column at its optimum feed stage. The product P concentration from the device is 98.7 mol% ethanol.

- 1) Find steam flow rate S , bottoms flow rate B , distillate flow rate D , product flow rate P , and recycle rate R - all in kmol/h - and find mole fraction of ethanol in recycle stream X_R .



Solving

$$F + S = P + B \quad (1)$$

$$Fz + Sx = PX_P + BX_B \quad (2)$$

$$B = 5S \quad (3)$$

3 equations & 3 unknowns

$$\left\{ \begin{array}{l} S = 201.89841 \text{ kmol/h} \\ B = 1009.49205 \text{ kmol/h} \\ P = 192.40636 \text{ kmol/h} \end{array} \right.$$

$$\begin{array}{l} S = 201.89841 \text{ kmol/h} \\ B = 1009.49205 \text{ kmol/h} \\ P = 192.40636 \text{ kmol/h} \\ D = 600.94921 \text{ kmol/h} \\ R = 408.543 \text{ kmol/h} \\ X_R = 0.564835 \end{array}$$

LMD analysis: $\bar{L} = B$ $\star$ $\bar{V} = S$ $\star$

$L' = \bar{L}$ $\star$ $V' = F + \bar{V}$ $\star$

$L' = R + L$ $\star$ $V = V'$ $\star$

$V = L + D$ $\star$

Now finding X_R & D

$$D = P + R \quad (1)$$

$$DX_D = PX_P + RX_R \quad (2)$$

$$\bar{V} = S$$

$$V' = F + S$$

$$V = F + S$$

$$L + D = F + S \text{ w/ } L = D$$

$$D = \frac{F+S}{2} \Rightarrow D = 600.949 \text{ kmol/h}$$

$$B = R + L$$

$$B = R + D \quad (3)$$

3 equations and three unknowns

$$600.949 = P + R$$

$$0.7(600.949) = 0.987P + RX_R$$

$$100.949205 = R + 600.949$$

$$\left\{ \begin{array}{l} R = 408.543 \text{ kmol/h} \\ P = 192.406 \text{ kmol/h} \\ X_R = 0.564835 \end{array} \right.$$

- 8) For the distillation column find the optimum feed plate for the recycle stream, the optimum feed plate for the fresh feed, and the total number of equilibrium stages required.

$$V_y = Lx + Dx_D$$

$\Downarrow$

$$\text{TOP Operating Line: } \star \left\{ y = \left(\frac{L}{V} \right) x + \left(\frac{D}{V} \right) x_D \right\}$$

$$V'y = L'x + PXP$$

$$\left\{ y = \left(\frac{L'}{V'} \right) x + \frac{PXP}{V'} \right\}$$

Middle Operating Line: *

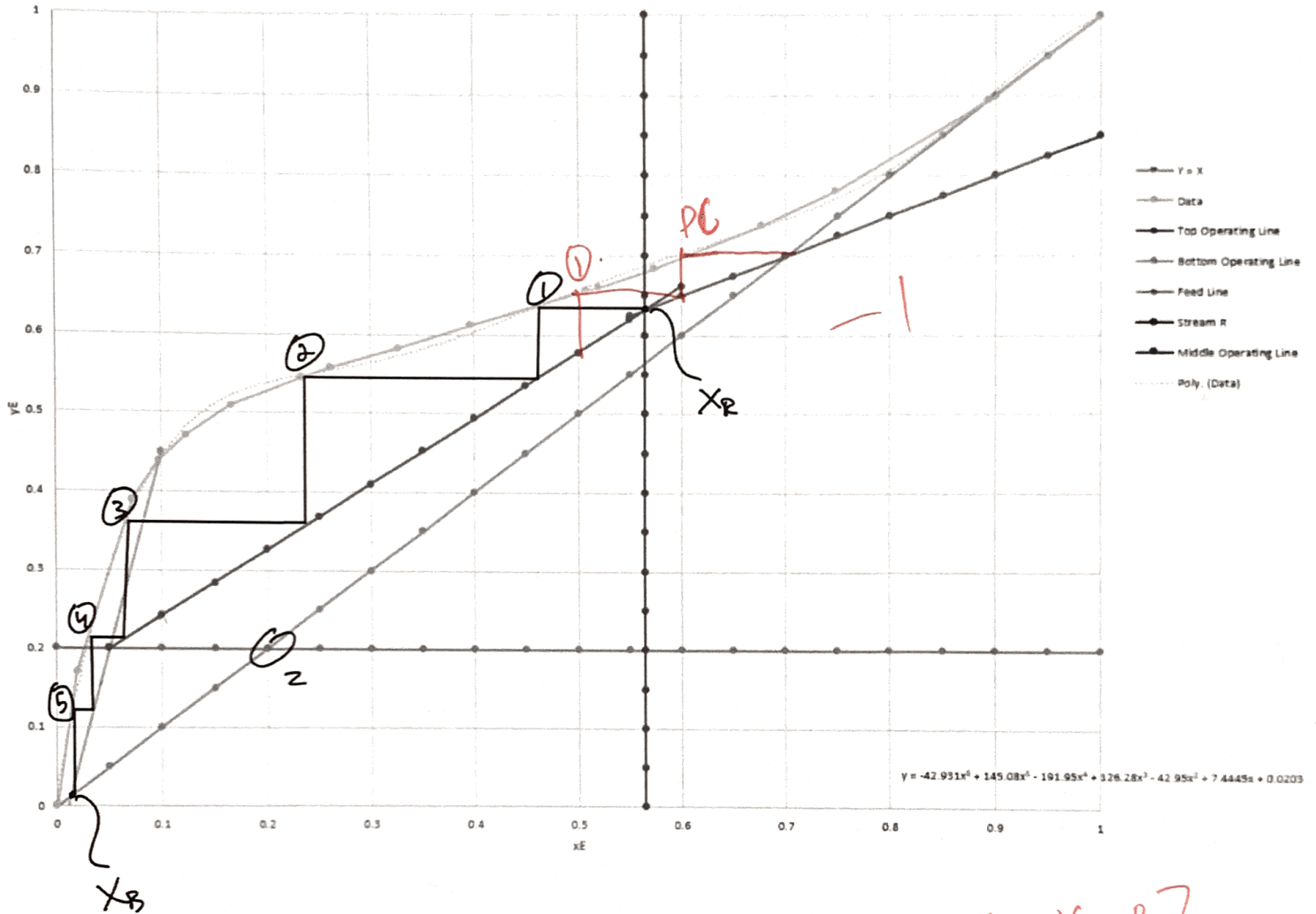
Line

$$\bar{V}y = \bar{L}x - (Bx_B - S_y)$$

$$\left\{ y = \left(\frac{\bar{L}}{\bar{V}} \right) x - \left(\frac{Bx_B - S_y}{\bar{V}} \right) \right\}$$

Bottom Operating Line: *

Line



5 Stages

Optimum feed plate for Fresh feed: Stage 1

Optimum feed plate for Recycle stream: Between Stage 4 & 5